

Code Containing README

Software Required

To run this project, the following software is required:

1. **Python 3.10 or above**
2. **Streamlit**
3. **A code editor** such as VS Code, PyCharm, or any text editor
4. **Web browser** to view the Streamlit interface

Project Files

The project folder contains the following main files:

- `app.py` – Main Streamlit application file
- `engine.py` – Contains the SQL conversion logic
- `dataset.json` – Knowledge base containing Oracle and PostgreSQL query pairs
- `pages/1_Converter.py` – Converter page for entering Oracle SQL and viewing PostgreSQL output
- `requirements.txt` – File used for storing required Python libraries

Steps to Run the Code

Step 1: Extract the project folder

Extract the zip file and open the extracted folder named:

```
sql_ui
```

Step 2: Open terminal or command prompt

Go to the project directory using:

```
cd sql_ui
```

Step 3: Install required library

Install Streamlit using the following command:

```
pip install streamlit
```

If a requirements file is used, then run:

```
pip install -r requirements.txt
```

Step 4: Run the application

Start the Streamlit application by running:

```
streamlit run app.py
```

Step 5: Open the web interface

After running the command, Streamlit will generate a local URL, usually:

```
http://localhost:8501
```

Open this link in a browser.

Step 6: Use the SQL Converter

Click on **Enter Converter**.

Enter an Oracle SQL query or select one of the sample queries.

Click on **Convert** to get the PostgreSQL query along with a confidence score.

Example Input

```
SELECT SYSDATE FROM dual;
```

Example Output

```
SELECT CURRENT_DATE;
```